

Math 142, S12
Exam 2

Section:

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	18	
2	24	
3	20	
4	20	
5	18	
Total	100	

Good Luck !

Problem 1. (18 points) Determine whether the sequence is convergent or divergent. If it is convergent, find its limit.

(a) $a_n = \frac{n^2 + n}{e^n}$

Let $f(x) = \frac{x^2 + x}{e^x}$

$$\begin{aligned} \lim_{x \rightarrow \infty} f(x) &= \lim_{x \rightarrow \infty} \frac{x^2 + x}{e^x} = \lim_{x \rightarrow \infty} \frac{2x + 1}{e^x} \\ &= \lim_{x \rightarrow \infty} \frac{2}{e^x} = 0 \end{aligned}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{n^2 + n}{e^n} = 0 \Rightarrow \{a_n\} \text{ Converges.}$$

(b) $a_n = \frac{2 + n^3}{5 - n^2 + 6n^3}$

$$\begin{aligned} \lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} \frac{2 + n^3}{5 - n^2 + 6n^3} = \lim_{n \rightarrow \infty} \frac{(2 + n^3)/n^3}{(5 - n^2 + 6n^3)/n^3} \\ &= \lim_{n \rightarrow \infty} \frac{\frac{2}{n^3} + 1}{\frac{5}{n^3} - \frac{1}{n} + 6} \end{aligned}$$

$$= \frac{1}{6} \Rightarrow \{a_n\} \text{ Converges.}$$

(c) $a_n = (-1)^n \cos\left(\frac{1}{n}\right)$

[Notice the change of sign of $(-1)^n \cos\left(\frac{1}{n}\right)$]

$$\lim_{n \rightarrow \infty} a_{2n} = \lim_{n \rightarrow \infty} (-1)^{2n} \cos\left(\frac{1}{2n}\right) = \cos 0 = 1$$

$$\lim_{n \rightarrow \infty} a_{2n+1} = \lim_{n \rightarrow \infty} (-1)^{2n+1} \cos\left(\frac{1}{2n+1}\right) = -\cos 0 = -1$$

$\Rightarrow \{a_n\}$ is divergent.

Problem 2. (24 points) Determine whether the series is convergent or divergent.

(a) $\sum_{n=1}^{+\infty} \frac{2^n}{3^n}$ (If convergent, also find its sum for this one)

$$\frac{2}{3} < 1 \Rightarrow \sum_{n=1}^{+\infty} \frac{2^n}{3^n} \text{ converges}$$

and

$$\sum_{n=1}^{+\infty} \frac{2^n}{3^n} = \frac{\frac{2}{3}}{1 - \frac{2}{3}} = \frac{\frac{2}{3}}{\frac{1}{3}} = 2$$

$$\hookrightarrow \sum_{n=1}^{+\infty} \frac{2}{3} \left(\frac{2}{3}\right)^{n-1}$$

(b) $\sum_{n=2}^{+\infty} \frac{1}{n(\ln n)^2}$

Let $f(x) = \frac{1}{x(\ln x)^2}$, clearly $f(x)$ decreases at $[2, +\infty)$.

$$\int_2^{+\infty} f(x) dx = \int_2^{+\infty} \frac{1}{x(\ln x)^2} dx = \lim_{t \rightarrow +\infty} \int_2^t \frac{1}{x(\ln x)^2} dx$$

$$\stackrel{u=\ln t}{=} \lim_{t \rightarrow +\infty} \int_{\ln 2}^{\ln t} \frac{1}{u^2} du$$

$$= \lim_{t \rightarrow +\infty} \left(-\frac{1}{u}\right) \Big|_{\ln 2}^{\ln t} = \lim_{t \rightarrow +\infty} \left(-\frac{1}{\ln t} + \frac{1}{\ln 2}\right) = \frac{1}{\ln 2}$$

$$\Rightarrow \sum_{n=2}^{+\infty} \frac{1}{n(\ln n)^2} \text{ Converges.}$$

(c) $\sum_{n=1}^{+\infty} \frac{\sqrt{n^2+2}}{n^2+n+1}$

(use the limit Comparison Test)

Let $a_n = \frac{\sqrt{n^2+2}}{n^2+n+1}$, $b_n = \frac{1}{n}$.

$$\lim_{n \rightarrow +\infty} \frac{a_n}{b_n} = \lim_{n \rightarrow +\infty} \frac{\frac{\sqrt{n^2+2}}{n^2+n+1}}{\frac{1}{n}} = \lim_{n \rightarrow +\infty} \frac{\sqrt{n^2+2} \cdot n}{n^2+n+1}$$

$$= \lim_{n \rightarrow +\infty} \frac{(\sqrt{n^2+2}) \cdot n/n^2}{(n^2+n+1)/n^2}$$

$$= \lim_{n \rightarrow +\infty} \frac{\sqrt{n^2+2}/n}{1 + \frac{1}{n} + \frac{1}{n^2}} = \lim_{n \rightarrow +\infty} \frac{\sqrt{1+\frac{2}{n^2}}}{1 + \frac{1}{n} + \frac{1}{n^2}} = \frac{\sqrt{1}}{1} = 1.$$

$$\sum_{n=1}^{+\infty} b_n = \sum_{n=1}^{+\infty} \frac{1}{n} \text{ is divergent} \Rightarrow \sum_{n=1}^{+\infty} a_n = \sum_{n=1}^{+\infty} \frac{\sqrt{n^2+2}}{n^2+n+1} \text{ is divergent.}$$

Problem 3. (20 points) Determine whether the series is conditionally convergent, absolutely convergent or divergent.

(a) $\sum_{n=2}^{+\infty} \frac{(-1)^n}{\ln n}$

Consider $\sum_{n=2}^{+\infty} \frac{(-1)^n}{\ln n}$. Let $b_n = \frac{1}{\ln n}$

i) clearly $b_{n+1} = \frac{1}{\ln(n+1)} < b_n = \frac{1}{\ln n}$

ii) $\lim_{n \rightarrow +\infty} \frac{1}{\ln n} = 0$

Thus, by the Alternating Series Test, $\sum_{n=2}^{+\infty} \frac{(-1)^n}{\ln n}$ is convergent.

Now, consider $\sum_{n=2}^{+\infty} \left| \frac{(-1)^n}{\ln n} \right| = \sum_{n=2}^{+\infty} \frac{1}{\ln n}$.

clearly $\frac{1}{\ln n} > \frac{1}{n}$ when $n \geq 2$

$\sum_{n=2}^{+\infty} \frac{1}{n}$ is divergent $\Rightarrow \sum_{n=2}^{+\infty} \frac{1}{\ln n}$ is divergent by the Comparison Test.

Thus, $\sum_{n=2}^{+\infty} \frac{(-1)^n}{\ln n}$ is conditionally convergent.

(b) $\sum_{n=1}^{+\infty} \frac{(-1)^n e^{\frac{1}{n}}}{n^2}$

Let $a_n = \frac{e^{\frac{1}{n}}}{n^2}$, $b_n = \frac{1}{n^2}$

$\lim_{n \rightarrow +\infty} \frac{a_n}{b_n} = \lim_{n \rightarrow +\infty} \frac{\frac{e^{\frac{1}{n}}}{n^2}}{\frac{1}{n^2}} = \lim_{n \rightarrow +\infty} \frac{e^{\frac{1}{n}}}{n^2} \cdot n^2 = \lim_{n \rightarrow +\infty} e^{\frac{1}{n}} = e^0 = 1$

Since $\sum_{n=1}^{+\infty} \frac{1}{n^2}$ is convergent, we get $\sum_{n=1}^{+\infty} \frac{e^{\frac{1}{n}}}{n^2}$ is also convergent

by the Limit Comparison Test.

$\Rightarrow \sum_{n=1}^{+\infty} \left| \frac{(-1)^n e^{\frac{1}{n}}}{n^2} \right| = \sum_{n=1}^{+\infty} \frac{e^{\frac{1}{n}}}{n^2}$ is convergent.

$\Rightarrow \sum_{n=1}^{+\infty} \frac{(-1)^n e^{\frac{1}{n}}}{n^2}$ is absolutely convergent.

Problem 4. (20 points) Find the radius of convergence and interval of convergence of the series.

(a) $\sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n 2^n}$

Let $a_n = \frac{(-1)^n x^n}{n 2^n}$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{\frac{(-1)^{n+1} x^{n+1}}{(n+1) 2^{n+1}}}{\frac{(-1)^n x^n}{n 2^n}} \right| = \left| \frac{x^{n+1}}{(n+1) 2^{n+1}} \cdot \frac{n 2^n}{x^n} \right| = \left| \frac{n}{n+1} \cdot \frac{x}{2} \right|$$

$$= \left| \frac{1}{1 + 1/n} \cdot \frac{x}{2} \right| \rightarrow \left| \frac{x}{2} \right| \text{ as } n \rightarrow \infty$$

$$\Rightarrow \left| \frac{x}{2} \right| < 1 \Rightarrow |x| < 2 \Rightarrow R = 2.$$

For $x=2$, we have $\sum_{n=1}^{\infty} \frac{(-1)^n 2^n}{n \cdot 2^n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ converges

$x=-2$, we have $\sum_{n=1}^{\infty} \frac{(-1)^n (-2)^n}{n \cdot 2^n} = \sum_{n=1}^{\infty} \frac{2^n}{n \cdot 2^n} = \sum_{n=1}^{\infty} \frac{1}{n}$ diverges

Thus the interval of convergence is $(-2, 2]$.

(b) $\sum_{n=0}^{+\infty} \sqrt{n} (x-1)^n$

Let $a_n = \sqrt{n} (x-1)^n$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{\sqrt{n+1} (x-1)^{n+1}}{\sqrt{n} (x-1)^n} \right| = \left| \sqrt{\frac{n+1}{n}} \cdot (x-1) \right| = \left| \sqrt{1 + \frac{1}{n}} (x-1) \right|$$

$$\rightarrow |x-1| \text{ as } n \rightarrow \infty$$

$$\Rightarrow |x-1| < 1 \Rightarrow R = 1.$$

$$|x-1| < 1 \Rightarrow 0 < x < 2.$$

For $x=0$, we have $\sum_{n=0}^{+\infty} \sqrt{n} (-1)^n$ diverges.

$x=2$, we have $\sum_{n=0}^{+\infty} \sqrt{n} (1)^n = \sum_{n=0}^{+\infty} \sqrt{n}$ diverges.

Thus the interval of convergence is $(0, 2)$.

Problem 5. (18 Points) Find the Maclaurin series (power series centered at 0) and its radius of convergence for the following functions.

(a) $f(x) = \frac{1}{2}(e^x + e^{-x})$

$$e^x = \sum_{n=0}^{+\infty} \frac{x^n}{n!} \quad e^{-x} = \sum_{n=0}^{+\infty} \frac{(-x)^n}{n!} = \sum_{n=0}^{+\infty} \frac{(-1)^n x^n}{n!}$$

$$f(x) = \frac{1}{2}(e^x + e^{-x}) = \frac{1}{2} \left(\sum_{n=0}^{+\infty} \frac{x^n}{n!} + \sum_{n=0}^{+\infty} \frac{(-1)^n x^n}{n!} \right)$$

$$= \sum_{n=0}^{+\infty} \frac{1}{2}(1 + (-1)^n) \frac{x^n}{n!} \rightarrow \frac{1}{2}(1 + (-1)^n) = \begin{cases} 1, & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$$

$$= \sum_{n=0}^{+\infty} \frac{x^{2n}}{(2n)!}$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{x^{2(n+1)}}{(2n+2)!} \cdot \frac{(2n)!}{x^{2n}} \right| = \left| \frac{x^2}{(2n+1)(2n+2)} \right| \rightarrow 0 \text{ as } n \rightarrow +\infty$$

$$\Rightarrow R = +\infty$$

(b) $f(x) = \frac{x^2}{1-2x}$

$$\frac{1}{1-2x} = \sum_{n=0}^{+\infty} (2x)^n = \sum_{n=0}^{+\infty} 2^n x^n$$

$$\Rightarrow f(x) = x^2 \cdot \sum_{n=0}^{+\infty} 2^n x^n$$

$$= \sum_{n=0}^{+\infty} 2^n x^{n+2}$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{2^{n+1} x^{n+3}}{2^n x^{n+2}} \right| = |2x|$$

$$|2x| < 1 \Rightarrow |x| < \frac{1}{2} \Rightarrow R = \frac{1}{2}$$

Extra Credit. (6 points) Find the Maclaurin series for the function

$$f(x) = \cos^2\left(\frac{x}{2}\right).$$

(Hint: Apply the half-angle identity first.)

Notice $\cos^2\left(\frac{x}{2}\right) = \frac{\cos x + 1}{2}$

$$\cos x = \sum_{n=0}^{+\infty} (-1)^n \frac{x^{2n}}{(2n)!}.$$

$$\Rightarrow \cos^2\left(\frac{x}{2}\right) = \frac{1}{2} \left[\left(\sum_{n=0}^{+\infty} (-1)^n \frac{x^{2n}}{(2n)!} \right) + 1 \right]$$

$$= \left(\sum_{n=0}^{+\infty} (-1)^n \frac{x^{2n}}{(2n)!} \right) + \frac{1}{2}$$

$$= 1 + \sum_{n=1}^{+\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

$$= 1 - \frac{x^2}{2 \cdot 2!} + \frac{x^4}{2 \cdot 4!} - \frac{x^6}{2 \cdot 6!} + \dots$$